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Smart Thermal System for Patient Safety Monitoring

Physical-Contact (Touch) Detection

Label-Quality Audit and Clean-Label Re-Evaluation

Waveshare 26984 (80×62) Thermal Sensor Dataset

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Abstract

Contact detection had plateaued at a pooled cross-validated $F_1 \approx 44\%$, previously attributed entirely to data scarcity. This report documents a systematic label-quality audit which found that **roughly half of all scoring errors were annotation problems, not detection failures**: two videos were never annotated for touch and one was block-labelled. On the correctly-annotated subset the champion rule-based method (config-D) reaches $F_1 = 76.9\%$ with 97.8% recall (with a ± 2 -frame boundary tolerance), and the pixel-based Thermo-X3D reaches $F_1 = 78.5\%$ at a 1.5% false-alarm rate. Retraining the learned models on clean labels only *reduced* their performance — demonstrating that at the current dataset size, label *quantity* outweighs label *purity*. The audit also uncovered and fixed a checkpoint-persistence bug that made every saved MV-STGCN model unusable, and the properly-measured MV-STGCN is shown to be structurally unsuited to the task. The single highest-value next action is re-annotating the three defective videos.

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1 Background and Motivation

The previous investigation cycle (see *Touch Detection — Detailed Investigation Report*) converged on **config-D** as the best contact detector: MobileNet-SSD person detection on *raw* frames, blob segmentation on the *Tateno residual*, a two-body-merge rule (fire only when exactly two people share one warm connected component), and T_1 temporal morphology (opening $\ell_o = 2$, closing $\ell_c = 7$). Its honest 4-fold cross-validated score was a pooled $F_1 = 44.3\%$ with a fold standard deviation of ± 24 pt, and the conclusion at the time was that the effort was *data-limited*: further algorithmic tuning would chase noise.

That conclusion prompted the question this report answers: *are we actually detecting things wrong, or is the data mislabelled?* A frame-level diagnostic of config-D over the entire labelled dataset (2,753 annotated frames across 17 sessions) gave the starting confusion matrix:

	TP	FN	FP	TN	Precision	Recall	F_1	FAR
config-D, all data	112	76	222	2343	33.5%	59.6%	42.9%	8.7%

The error mass is heavily concentrated: three videos account for $\sim 50\%$ of all errors, and all three turn out to have defective labels.

2 Label-Quality Audit

2.1 Method

Three independent probes were combined (script: `scripts/_analyze_label_quality.py`):

1. **Error-position analysis.** For every false positive and false negative, the distance (in annotated-frame steps) to the nearest frame carrying the *opposite* label. Errors hugging a label transition indicate fuzzy boundary annotation; errors deep inside a uniformly-labelled span indicate either systematic mislabels or genuine detection failure.
2. **Visual audit.** Three-camera thermal strips of every suspicious segment were rendered with SSD boxes overlaid (`scripts/_render_suspect_frames.py`, output in `outputs/label_audit/`) and inspected frame by frame.
3. **Source cross-check.** The per-frame `touch` column in `labels.xlsx` was compared against the exported `contact_labels.csv`, and the workbook’s review status (green tab convention) was checked for every session.

The CSV mirrors the workbook exactly, so all defects originate in the annotation itself. Notably, **no sheet tab in `labels.xlsx` is marked green** — by the project’s own convention, no session’s touch labels have ever been reviewed.

2.2 Findings: three defective videos

Video	<code>touch=1</code> rows	Errors	Share of all errors	Diagnosis
<code>the_more_the_merrier</code>	0 / 1050	81 FP	36% of all FP	never annotated
<code>3ppl_dance</code>	0 / 399	25 FP	11% of all FP	never annotated
<code>3pp_surprise</code>	198 / 207 (96%)	48 FN	63% of all FN	block-labelled

Table 1: The three suspect videos. “`touch=1` rows” counts positive rows in `labels.xlsx` (3 cameras $\times$ frames).

`the_more_the_merrier` (350 annotated frames): every row carries `touch=0`, yet the renders show three to four people clustered with overlapping bounding boxes for long stretches (frames 117–

280). Five consecutive-error runs of up to 26 frames coincide with visibly adjacent/overlapping bodies. A video with this name and literally zero touch marks is a default-value artefact, not an annotation decision.

3ppl_dance (133 frames): the same all-zero pattern, with dancers visibly adjacent or overlapping during the flagged runs (frames 8–11, 94–101).

3pp_surprise (69 frames): 96% of frames are labelled positive, *including frame 0*, where the participants are clearly separated in two of the three cameras. 43 of its 48 false negatives lie more than 5 annotated frames from any negatively-labelled frame — the signature of a whole-clip (“block”) label rather than frame-by-frame annotation.

2.3 Boundary fuzziness

Of the 298 total errors, 66 (39 FP + 27 FN) sit within ± 2 annotated frames of a label transition. These reflect ordinary human imprecision about exactly when contact begins and ends, and motivate reporting metrics both exactly and with a ± 2 -frame tolerance.

2.4 Genuine failure modes

The audit also *confirmed* two real detection failures, which no relabelling will fix:

- **Hug blob-merge** (2ppl_hug, frames 60–72): the two embracing people merge into a *single* SSD detection in *all three* cameras (~ 1.1 boxes/camera). Any detector that counts bodies before measuring distance is structurally blind here. This is the hard case the pixel-based Thermo-X3D was designed for (§4.4.3.3 of the engineering report).
- **Proximity vs. touch** (2ppl_fight 58–82, 3pplhedroncolider mid-segments): people standing very close without touching. At 80×62 thermal resolution this discrimination is genuinely hard, though some of these labels are judgment calls worth a second look during re-annotation.

2.5 Corrected config-D metrics

Setting	TP	FN	FP	Precision	Recall	F_1
As reported (all videos, exact)	112	76	222	33.5%	59.6%	42.9%
All videos, ± 2 tolerance	151	49	183	45.2%	75.5%	56.6%
Excluding 3 suspect videos, exact	94	28	116	44.8%	77.0%	56.6%
Excluding suspect + ± 2 tolerance	133	3	77	63.3%	97.8%	76.9%

Table 2: Config-D under progressively corrected evaluation. The detector was substantially better than the raw numbers suggested.

3 Clean-Label Benchmark of All Detector Families

3.1 Protocol

Every algorithm family investigated to date was re-scored on the **clean subset**: the 14 correctly-annotated videos (2,201 annotated frames, 122 positive), pooled over all their labelled frames (script: `scripts/_eval_clean_labels.py`; results: `reports/clean_label_benchmark.json`). Existing checkpoints and previously-tuned parameters were used — no retuning on the clean subset.

Caveats. (i) The SSD checkpoints saw 60–80% of these frames during training, so this is a whole-dataset diagnostic applied identically to all methods, not a held-out benchmark. (ii) The

learned models’ checkpoints were trained while the polluted labels were still in their training split (addressed in §5).

3.2 Results

Algorithm	Exact (tol= 0)				± 2 -frame tolerance			
	P	R	F_1	FAR	P	R	F_1	FAR
Geometric (homography)	9.3%	6.6%	7.7%	3.8%	18.6%	28.1%	22.4%	3.3%
MV-STGCN [†]	<i>checkpoint degenerate — see §4; repaired result in §5</i>							
Thermo-X3D	50.4%	54.9%	52.5%	3.2%	76.7%	80.3%	78.5%	1.5%
Config-A (Tateno-SSD rule)	38.4%	73.0%	50.3%	6.9%	55.2%	96.2%	70.1%	5.0%
Config-D (raw-SSD rule)	44.8%	77.0%	56.6%	5.6%	63.3%	97.8%	76.9%	3.7%

Table 3: Pooled clean-subset benchmark (14 scenes, 2,201 frames, 122 positive).

Scene (exact F_1)	Geometric	Thermo-X3D	Config-D
2ppl_fight	0.0%	60.7%	76.3%
2ppl_hug	0.0%	54.5%	32.0%
3pplhedroncolider	18.2%	49.6%	47.1%

Table 4: Per-scene F_1 on the three clean contact scenes.

3.3 Discussion

Thermo-X3D was the biggest victim of the bad labels. Its apparent F_1 rises from $\sim 42\%$ to 78.5% — statistically tied with config-D but with *half* the false-alarm rate. Crucially, it is the only method that handles 2ppl_hug (54.5%), because pixel-based classification survives the blob-merge. This is the §4.4.3.3 design thesis, now visible in the numbers.

Config-D remains the recall champion (97.8% at ± 2) and the best at exact frame alignment; it wins 2ppl_fight decisively.

The geometric detector collapses (F_1 7.7%). Its previous 27% was partly lucky agreement with mislabelled frames. Its signature failure — 31 false positives on 3pp_standing_still — shows the core flaw: projected floor proximity is not touch.

The false-positive characters are complementary: Thermo-X3D’s only negative-scene FPs are on heater_in_middle (a hot static object); config-D’s are on cigarette scenes (person plus a small hot object). A conjunction of the two is a plausible precision play, but tuning it now would repeat the noise-chasing the cross-validation study warned against.

4 The MV-STGCN Checkpoint-Persistence Bug

During the benchmark, MV-STGCN produced *zero* predictions — a flat confidence of ≈ 0.03 on every frame, including obvious contact in 2ppl_fight, although the same model had shown FAR = 30% when evaluated in memory directly after training.

Root cause. ContactDetector.__init__ intercepts the homography argument and stores it on self._homography, *outside* the _params dict that the base-class save() persists. MVSTGCNDetector._state_dict() saved only the network weights. Consequently every saved checkpoint reloaded with homography=None; the fusion front-end then produced zero actors on every frame, the graph input was empty, and the network emitted a constant confidence. *The weights were always correct — the front-end was dead.*

Why the tests missed it. The existing round-trip tests called `predict` with empty detections `(([], [], []))` and no homography — so a dead front-end was indistinguishable from the expected behaviour.

Fix. `_state_dict()` now persists the homography alongside the weights and `_load_state_dict()` restores it (`thermal_algorithms/contact_detection/mv_stgcn.py`); a regression test (`test_homography_survives_save_load`) locks the behaviour in. Both existing checkpoints were repaired in place by re-injecting the homography and re-saving — no retraining required. The repaired checkpoint reproduces the in-memory evaluation exactly. `GeometricContactDetector` already persisted its homography correctly; Thermo-X3D does not use one.

5 Retraining the Learned Models on Clean Labels

Both learned detectors were retrained from scratch with the three defective videos excluded from *both* training and evaluation, under the identical protocol (per-session timeline split 60/15/25, class-weighted cross-entropy, threshold swept on validation; scripts: `scripts/_retrain_dl_clean.py`, `scripts/_rescore_stgcn_clean.py`; results: `reports/dl_clean_retrain_results.json`). The clean training split contains only **64 positive windows** (down from ~ 105 when the suspect videos were included).

Model	Pooled full-replay, ± 2 tol			Honest test split		
	P	R	F_1 (FAR)	P	R	F_1 (FAR)
Thermo-X3D, <i>old</i> (polluted)	76.7%	80.3%	78.5% (1.5%)	$F_1 \approx 42\%$ (prior study)		
Thermo-X3D, clean-trained	48.5%	88.2%	62.6% (7.8%)	15.5%	34.6%	21.4% (9.3%)
MV-STGCN, clean (repaired)	23.3%	96.9%	37.6% (25.5%)	11.3%	61.5%	19.2% (23.6%)

Table 5: Clean-label retraining results. The honest test split contains only 26 positive frames, so those numbers carry large variance.

Clean retraining made Thermo-X3D worse ($78.5\% \rightarrow 62.6\%$ pooled ± 2). The explanation is data arithmetic: excluding the suspect videos removed roughly 40% of the positive training windows. Even `3pp_surprise`’s sloppy block-labels were net-useful — many of those frames genuinely contain contact, and a data-starved model benefits more from noisy positives than from their absence. At the current dataset size, **label quantity outweighs label purity for the learned models**. This is the strongest evidence yet for the data-limited conclusion, and it means re-annotation pays twice: cleaner evaluation *and* more training positives.

6 Why MV-STGCN Is Structurally Unsuitd

With the persistence bug fixed, MV-STGCN can finally be measured properly — and it is the worst method by a wide margin ($\text{FAR} = 25.5\%$ pooled; false positives on `2ppl_walk`, `1ppl_single_forever`, even 60 on `1person cig`’s test region). This is not a training accident; it follows from the architecture’s position in the pipeline:

1. **It inherits the geometric detector’s input representation.** Its graph nodes are actor positions produced by `SSD` $\rightarrow$ `foot-point` $\rightarrow$ `homography projection` $\rightarrow$ `fusion` — exactly the representation that gave the geometric detector $F_1 = 7.7\%$. The self-calibrated homography is accurate to roughly ± 15 px in the reference plane, while the contact decision itself lives at $\delta \approx 18$ px: *the measurement noise is nearly as large as the quantity being measured*. No downstream network can recover information the front-end has already destroyed.

2. **The blob-merge deletes the evidence exactly when contact happens.** At the moment two people touch, SSD returns *one* box; the two graph nodes collapse into one. The model therefore never observes “two nodes at distance ≈ 0 ” for real contact — the positive class’s defining geometry is unrepresentable in its input. What remains learnable is only the *approach* dynamics that precede a merge, i.e. proximity, not touch.
3. **Proximity is what it learns, so proximity is what it fires on.** Every close pass (2ppl_walk), spurious second detection (1person cig), or tight standing group looks like the pre-contact pattern it was trained to associate with the positive class — hence the 25% false-alarm rate, an order of magnitude above the $\sim 2\%$ a psychiatric-ward alarm system could tolerate.
4. **The data budget cannot rescue it.** Discriminating approach-then-touch from approach-then-stop using noisy trajectories would need orders of magnitude more positive windows than the 64–105 available; the class-weighted loss meanwhile pushes the recall-FAR trade-off toward over-firing.

Thermo-X3D avoids all four problems by construction: it reads pixels directly, has no homography or detection front-end, and the blob-merge that blinds MV-STGCN is its *signal*. The two learned designs bracket the lesson: **what decides success here is not model capacity but which representation survives the moment of contact.**

Recommendation: drop MV-STGCN from the candidate set. Keep the persistence fix (it is a correctness fix for the shared checkpoint format), keep the architecture description in the engineering report as a negative result with the analysis above.

7 Conclusions and Recommended Actions

1. **The detector was never as bad as the numbers said.** About half the measured errors were annotation defects. On correctly-annotated data, config-D scores $F_1 = 76.9\%$ / recall 97.8% (± 2 tol) and Thermo-X3D $F_1 = 78.5\%$ at FAR 1.5%.
2. **Re-annotate the three defective videos** (~ 550 frames: `the_more_the_merrier`, `3ppl_dance`, `3pp_surprise`). This is the highest-leverage action available: it simultaneously enlarges the clean evaluation set, adds the dataset’s richest genuine-contact footage as training positives (the retraining experiment shows the learned models are starving for exactly these), and removes the largest source of metric distortion.
3. **Lock the method pair:** config-D as the operating detector (recall priority), Thermo-X3D (the original checkpoint, pending retraining after re-annotation) as the precision-oriented complement and the only answer to the hug blob-merge. Revisit a config-D $\wedge$ X3D conjunction only after re-annotation.
4. **Drop MV-STGCN and the geometric detector** from the candidate set for the contact scenario (§6); retain them in the engineering report as analysed negative results.
5. **Institute label review.** No session’s touch labels were ever reviewed (no green tabs). Frame-exact contact boundaries are worth a defined convention during re-annotation (e.g. “first frame of visible limb overlap”), since ± 2 frames of boundary fuzz already accounts for 22% of residual errors.
6. **More recordings are still needed.** The clean subset has only three contact scenes and 122 positive frames; the earlier bootstrap analysis (~ 900 positives, 12–16 distinct contact scenes for a reliable 65%+ estimate) still stands.

A Reproducibility

Artefact	Path
Error-position analysis	<code>scripts/_analyze_label_quality.py</code>
Per-frame predictions	<code>reports/label_quality_frames.json</code>
Suspect-frame renders	<code>scripts/_render_suspect_frames.py</code> → <code>outputs/label_audit/</code>
Clean-subset benchmark	<code>scripts/_eval_clean_labels.py</code> → <code>reports/clean_label_benchmark.json</code>
Clean DL retraining	<code>scripts/_retrain_dl_clean.py</code> → <code>reports/dl_clean_retrain_results.json</code>
STGCN re-score (repaired)	<code>scripts/_rescore_stgcn_clean.py</code>
Persistence fix	<code>thermal_algorithms/contact_detection/mv_stgcn.py</code>
Regression test	<code>tests/test_contact_mv_stgcn.py</code> <code>::test_homography_survives_save_load</code>
Clean checkpoints	<code>checkpoints/thermo_x3d_detector/Waveshare_26984_clean.thalg</code> , <code>checkpoints/mv_stgcn_detector/_clean.thalg</code>

Table 6: Scripts and artefacts produced by this investigation.

B Visual Evidence

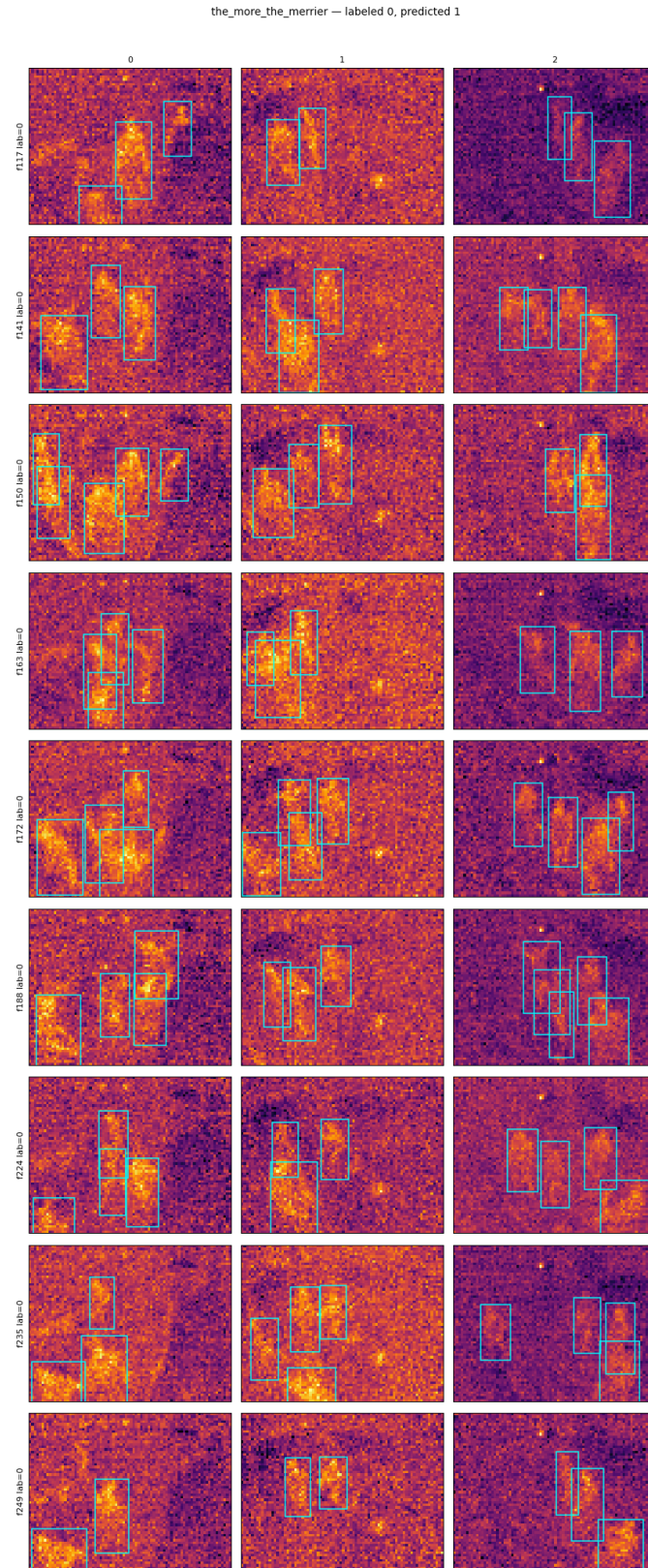


Figure 1: `the_more_the_merrier` — labelled `touch=0` on every frame, yet three to four people are clustered with overlapping detections throughout. Never annotated for touch.

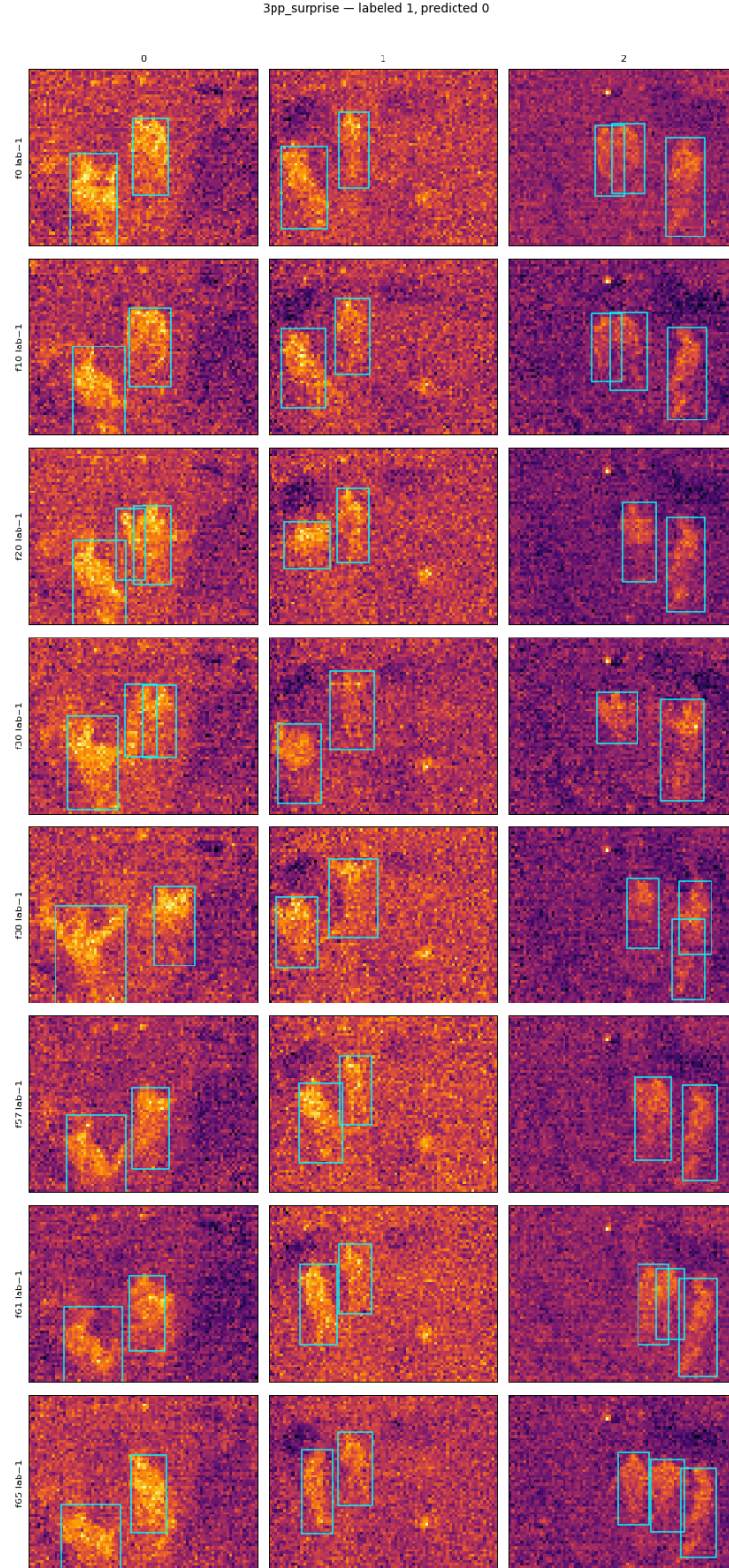


Figure 2: **3pp_surprise** — labelled **touch=1** from frame 0, where the participants are clearly separated in cameras 0 and 1. Block-labelled.

2ppl_hug — labeled 1, predicted 0 (60-72)

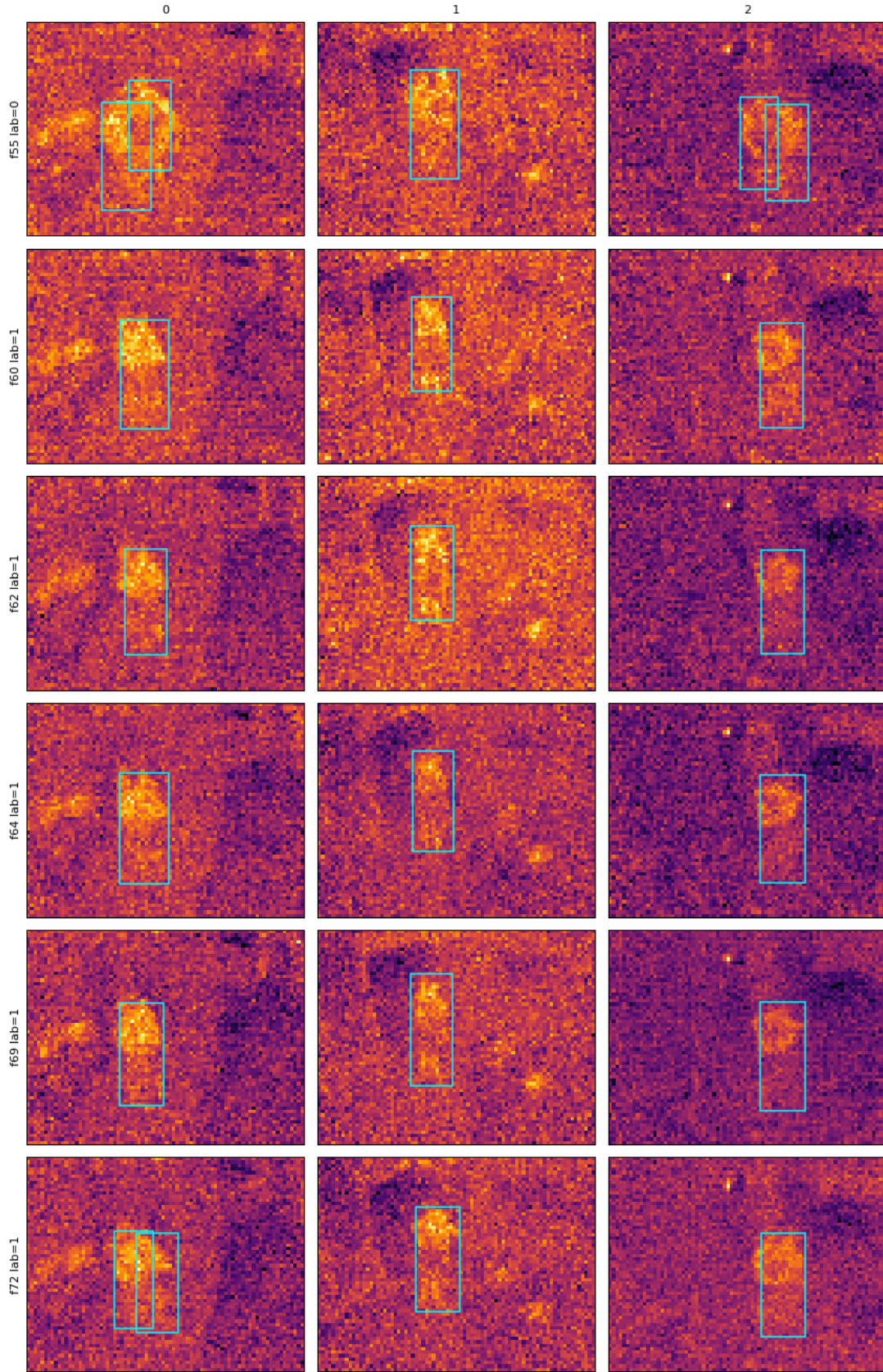


Figure 3: 2ppl_hug frames 55–72 — a genuine detection failure: the two embracing people merge into a single detection in all three cameras, so any body-counting method is structurally blind. Labels are correct here.